

Wildfire Raises Cloud-Forming Particles

The result

Wildfire reaching this remote ocean site **increases cloud-forming particles (CCN) by about 15–20%** during wildfire periods — roughly **+40 particles per cm^3** . The whole distribution shifts up and the high end gets fatter, so wildfire also makes **very high-particle episodes more common** (Figure 1). The finding holds across three different model types, each estimating between +10% and +24%.

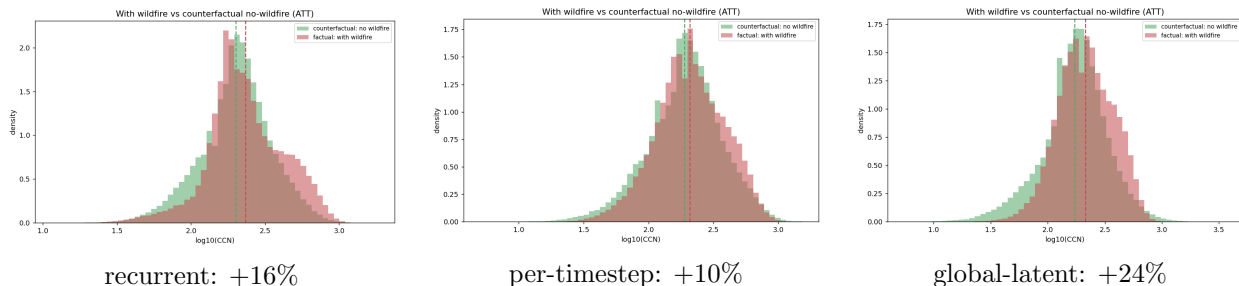


Figure 1: CCN during wildfire periods (red) vs. what it would have been without wildfire (green), for all three model types. Every one shows wildfire shifting the distribution to higher CCN (dashed lines = averages) with a fatter high-particle tail. Horizontal axis is a log scale. See “How we measured it” for exactly how these two curves are produced.

How we measured it

We trained one model on wildfire periods and one on clean periods. To separate wildfire’s own effect from the weather that carried it, we compare the two models *at the same weather*. Concretely, we take the weather from the **held-out wildfire periods** (real wildfire events the models did not train on) and run **both** models on it:

- the **red** curve is the *wildfire* model’s predicted CCN for that weather (“with wildfire”), and
- the **green** curve is the *clean* model’s prediction for the *same* weather (“what CCN would have been without wildfire”).

Both curves use the same input (wildfire-period weather); only the model differs. The gap between them is the wildfire effect. (We do not use the clean-period weather here — that would answer a different question.)

The three model types

All three make the LSTM output a *range* of possible CCN values (rather than a single guess) by injecting Gaussian noise; they differ in where the noise enters and how it varies across the trajectory:

- **Per-timestep:** a standard LSTM with a random noise vector concatenated onto the weather inputs, redrawn *fresh and independent at every timestep*.
- **Global-latent:** the *same* setup, except the noise vector is drawn *once per trajectory* and held fixed across all timesteps (one shared “latent code”).
- **Recurrent:** unlike the other two, the noise is *added to* the LSTM’s *hidden state* (not concatenated to the inputs) at each step, so the randomness *accumulates* through the sequence.

What to keep in mind

This is the effect during *real* wildfire periods. It assumes the weather data captures everything that differs between wildfire and clean periods, and we have not yet attached formal error bars — the estimate ranges from +10% to +24% across model types.